

Q.17 — Deuteric Alteration of Ca-Rich Plagioclase

DMG 2019 · Igneous Petrology: Deuteric / Late-Magmatic Alteration · Reference: Perkins, Mineralogy, 3rd ed. (supplemented with standard petrology concepts)

- Q.17.** Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?
- (a) Saussuritization
 - (b) Devitrification
 - (c) Chloritization
 - (d) Uralitization

A note on sourcing: This question belongs to igneous petrology (deuteric/late-magmatic alteration processes), which sits outside the main scope of the uploaded *Mineralogy* textbook — Perkins touches it only briefly, under the mineral epidote's entry. The core textbook citation below is genuine and page-cited; the broader classification of the four alteration processes is standard igneous petrology knowledge (as covered more fully in a dedicated text such as Winter's *Igneous and Metamorphic Petrology*).

1. The Underlying Concept — What Is Deuteric Alteration?

Deuteric alteration (also called autometamorphism or autometasomatism) is the alteration of an igneous rock's primary minerals by the **magma's own residual fluids**, during the last, waning stages of crystallization — while the body is still cooling, before it becomes a fully solid, cold rock. This distinguishes it from ordinary hydrothermal alteration, which acts later, from external fluids moving through an already-solidified rock.

As a magma crystallizes, water and other volatile components that don't fit into the early-forming silicate minerals become concentrated in the last residual melt/fluid. This fluid reacts with minerals that already crystallized, converting them into new, typically hydrous, lower-temperature mineral assemblages, all while the system is still essentially self-contained ("auto-").

2. The Four Named Deuteric Alteration Processes

Process	What Alters	Product
Saussuritization	Ca-rich plagioclase (labradorite, bytownite, anorthite)	Fine-grained aggregate of albite + epidote/zoisite/clinozoisite (± sericite, calcite) — collectively "saussurite"
Devitrification	Volcanic glass	Fine-grained crystalline quartz + feldspar (spherulites, etc.) — unrelated to plagioclase alteration
Chloritization	Mafic minerals (biotite, hornblende, pyroxene)	Chlorite
Uralitization	Pyroxene (commonly augite)	Fibrous/amphibole pseudomorphs after pyroxene, called " uralite "

3. Textbook Confirmation — the Product Side of the Reaction

Perkins' *Mineralogy* directly confirms the product of this alteration, under the mineral epidote's *Varieties* section:

"Saussurite is a name given to fine-grained epidote produced by alteration of plagioclase."
— Perkins, *Mineralogy*, p. 391 (Epidote, Varieties)

The book also illustrates this exact style of alteration with a photograph:

"FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar."
— Perkins, *Mineralogy*, p. 391

octahedra. Small amounts of Ti, Fe, Mg, Na, and K may be present, but no major solid solutions are known.

Occurrences and Associations

Occurrence and Associations

Lawsonite is a metamorphic mineral typical of the blueschist facies. Associated high-pressure minerals include glaucophane, jadeite, pumpellyite, or aragonite. Other associated minerals include chlorite, plagioclase, titanite, quartz, and epidote.

Related Minerals

Hemimorphite, $\text{Zn}_4(\text{Si}_2\text{O}_7)(\text{OH})_2 \cdot \text{H}_2\text{O}$, and ilvaite, $\text{CaFe}_3\text{O}(\text{Si}_2\text{O}_7)(\text{OH})$, are other paired tetrahedral silicates in which all SiO_4 tetrahedra are paired. Other minerals usually considered in the same group are epidote, $\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$; clinozoisite, $\text{Ca}_2\text{Al}_3\text{Si}_3\text{O}_{12}(\text{OH})$; piemontite, $\text{Ca}_2(\text{Al,Mn})_3\text{Si}_3\text{O}_{12}(\text{OH})$; allanite, $(\text{Ca,Ce})_2(\text{Al,Fe,Mg})_3\text{Si}_3\text{O}_{12}(\text{OH})$; and vesuvianite, $\text{Ca}_{10}(\text{Mg,Fe})_2\text{Al}_4\text{Si}_9\text{O}_{34}(\text{OH,F})_4$.

Epidote

$\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$

Origin of Name

From the Greek word *epididonai*, meaning “increase,” referring to the base of an epidote prism, one side of which is longer than the other.

Hand Specimen Identification

Epidote is characterized by its pistachio-green color, habit, one perfect cleavage, and association. It is often fine grained and massive (Figure 19).

Physical Properties

hardness	6 to 7
specific gravity	3.4 to 3.5
cleavage/fracture	perfect (001), poor (100)/uneven
luster/transparency	vitreous/transparent to translucent
color	pistachio-green, yellow-green to black
streak	white

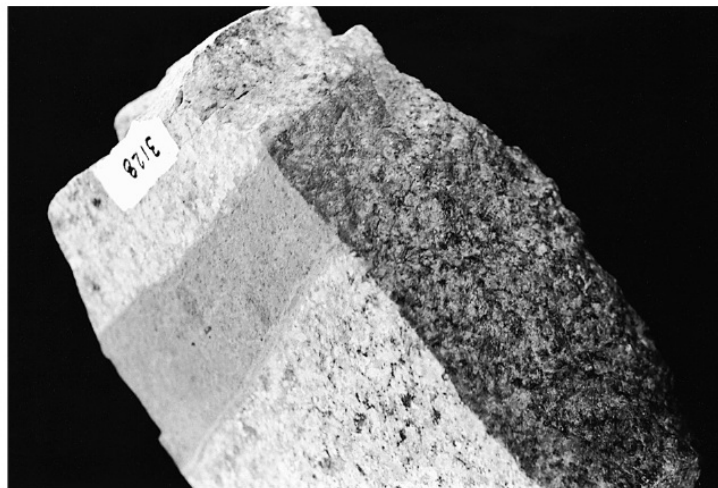


FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar.

The Epidote entry (Perkins, *Mineralogy*, p. 391), including Figure 19 — a vein of epidote in granite formed by feldspar alteration — and the Varieties note defining saussurite as fine-grained epidote produced by alteration of plagioclase.

Why the reaction yields albite + epidote specifically

Plagioclase is a solid solution between albite ($\text{NaAlSi}_3\text{O}_8$) and anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$). During saussuritization:

Anorthite component + H_2O ($\pm \text{CO}_2$) $\rightarrow$ Epidote/Zoisite + (Al, Ca redistributed) | Albite component largely unaltered

The **Ca-Al-rich (anorthite) component** breaks down, releasing Ca and Al that recombine with the residual fluid to form epidote-group minerals, while the **Na-rich (albite) component** survives essentially untouched. This is exactly why the reaction is described as producing "almost pure albite **and** epidote" — both phases persist side by side as breakdown products.

4. Evaluating Each Option

Option	Reasoning	Verdict
(a) Saussuritization	Defining reaction is exactly $\text{Ca-plagioclase} \rightarrow \text{albite} + \text{epidote/zoisite}$; confirmed by Perkins' saussurite entry	Correct
(b) Devitrification	Applies to volcanic glass turning crystalline — no plagioclase or albite/epidote relationship at all	Eliminated
(c) Chloritization	Produces chlorite from mafic (Fe-Mg) minerals, not albite + epidote from feldspar	Eliminated
(d) Uralitization	Affects pyroxene, producing amphibole (uralite) — unrelated to feldspar breakdown	Eliminated

✓ Answer: (a) Saussuritization

5. Probable Follow-Up Exam Questions

Q. What is "saussurite"?

A. A fine-grained, greenish, massive aggregate of albite, epidote/zoisite, sericite, and calcite that replaces Ca-rich plagioclase; classically seen in "saussuritized gabbro."

Q. What does uralitization produce, and from what mineral?

A. It converts pyroxene (typically augite) into a fibrous amphibole pseudomorph called uralite.

Q. Is devitrification an alteration of a crystalline mineral?

A. No — it specifically refers to crystallization of previously glassy (amorphous) volcanic material, not alteration of an existing crystal.

Q. Which rock type most classically shows saussuritized plagioclase?

A. Gabbro and diorite — "saussuritized gabbro" is the standard textbook example.

Q. How does deuteric alteration differ from hydrothermal alteration?

A. Deuteric alteration is caused by the magma's own residual fluid during the final stages of crystallization (autometamorphic); hydrothermal alteration is caused by external fluids acting on an already-solidified rock, often much later.